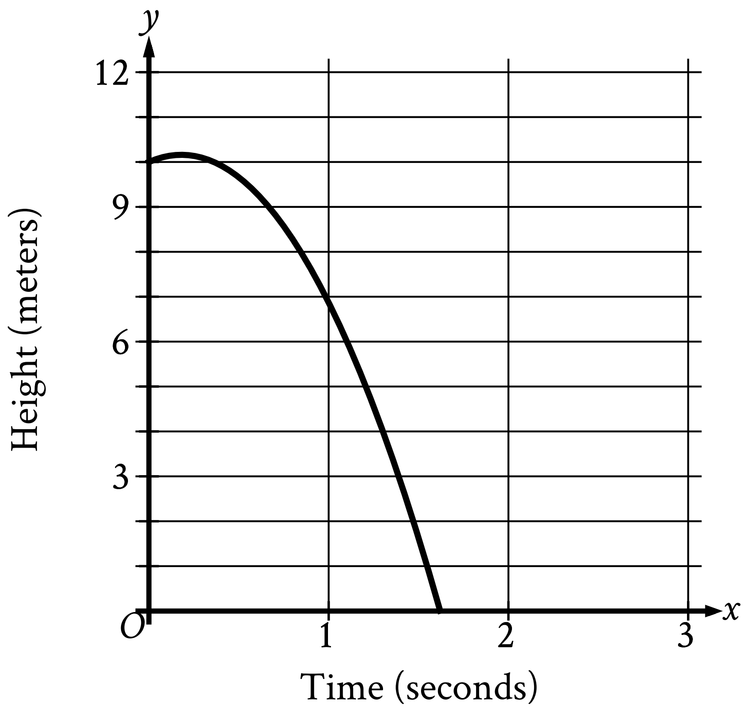
# Question ID: 9ff88bb5

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Advanced Math | Nonlinear functions | Medium |

**Question**




A competitive diver dives from a platform into the water. The graph shown gives the height above the water $y$, in meters, of the diver $x$ seconds after diving from the platform. What is the best interpretation of the *x*-intercept of the graph?

**Answer**

A. The diver reaches a maximum height above the water at $1.6$ seconds.

B. The diver hits the water at $1.6$ seconds.

C. The diver reaches a maximum height above the water at $0.2$ seconds.

D. The diver hits the water at $0.2$ seconds.

**Correct Answer: B**

**Rationale**

Choice B is correct. It's given that the graph shows the height above the water $y$, in meters, of a diver $x$ seconds after diving from a platform. The *x*-intercept of a graph is the point at which the graph intersects the *x*-axis, or when the value of $y$ is $0$. The graph shown intersects the *x*-axis between $x = 1$ and $x = 2$. In other words, the diver is $0$ meters above the water, or hits the water, between $1$ and $2$ seconds after diving from the platform. Of the given choices, only choice B includes an interpretation where the diver hits the water between $1$ and $2$ seconds. Therefore, the best interpretation of the *x*-intercept of the graph is the diver hits the water at $1.6$ seconds.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. This is the best interpretation of the maximum value, not the *x*-intercept, of the graph.

Choice D is incorrect and may result from conceptual errors.